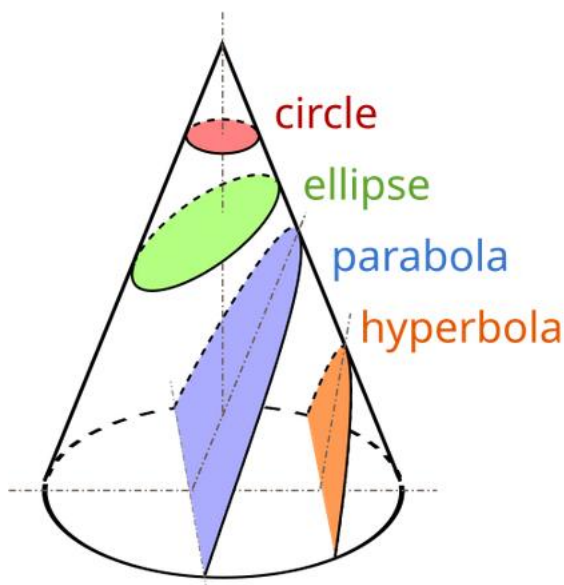


General Equation of Second Degree

Conics

A conic section, conic, or a quadratic curve is a curve obtained from a cone's surface intersecting a plane. The three types of conic sections are the hyperbola, the parabola, and the ellipse; the circle is a special case of the ellipse, though it was sometimes considered a fourth type.



Identification:

The general equation of the 2nd degree is,

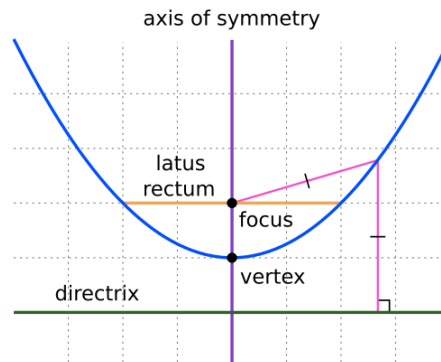
$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \dots\dots\dots (1)$$

If $\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$ then (1) represents

- (a) Circle if $\Delta \neq 0$ and $a = b, h = 0$
- (b) Parabola if $\Delta \neq 0$ and $ab - h^2 = 0$
- (c) Ellipse if $\Delta \neq 0$ and $ab - h^2 > 0$
- (d) Hyperbola if $\Delta \neq 0$ and $ab - h^2 < 0$

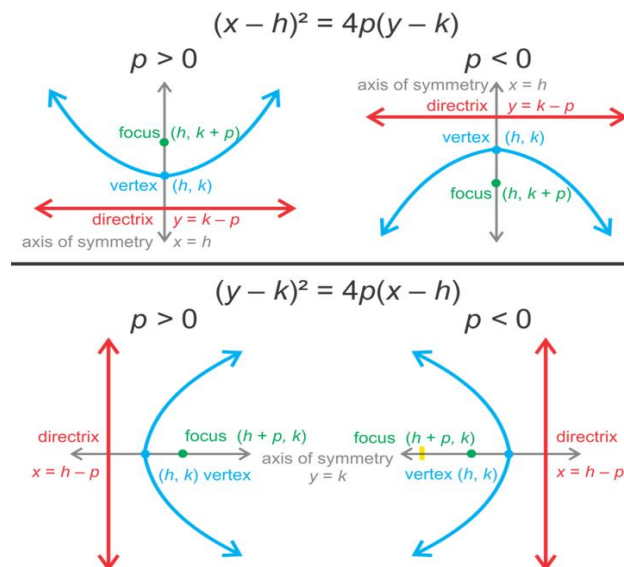
Parabola:

A parabola is a curve in which every point on the curve is equidistant from a point called a focus and a straight line called a directrix. It is a U-shaped curve that appears often in mathematics.



The general equation of a parabola is often expressed as $(x - h)^2 = 4p(y - k)$ or $(y - k)^2 = 4p(x - h)$

Where (h, k) is the vertex.



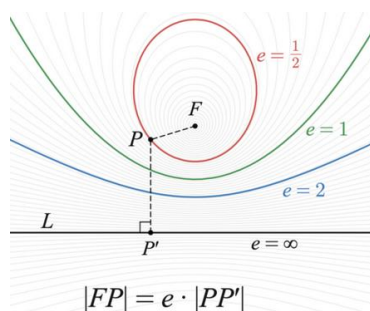
The standard equation of a regular parabola is $y^2 = 4ax$.

Forms of Parabolas

Forms:	$y^2 = 4ax$	$y^2 = -4ax$	$x^2 = 4ay$	$x^2 = -4ay$
Equation of the axis:	$y = 0$	$y = 0$	$x = 0$	$x = 0$
Equation of the directrix:	$x = -a$	$x = a$	$y = -a$	$y = a$
Vertex:	$(0, 0)$	$(0, 0)$	$(0, 0)$	$(0, 0)$
Focus:	$(a, 0)$	$(-a, 0)$	$(0, a)$	$(0, -a)$
Length of latus rectum:	$4a$	$4a$	$4a$	$4a$

Eccentricity

It is the locus of all points P whose distance to a fixed point F (called the *focus*) is a constant multiple e (called the *eccentricity*) of the distance from P to a fixed line L (called the *directrix*).



We obtain, Parabola for $e = 1$, Ellipse for $0 < e < 1$, Hyperbola for $e > 1$.

For conics in standard position, these parameters have the following values, taking $a, b > 0$

conic section	equation	eccentricity (e)	linear eccentricity (c)
circle	$x^2 + y^2 = a^2$	0	0
ellipse	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	$\sqrt{1 - \frac{b^2}{a^2}}$	$\sqrt{a^2 - b^2}$
parabola	$y^2 = 4ax$	1	N/A
hyperbola	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$\sqrt{1 + \frac{b^2}{a^2}}$	$\sqrt{a^2 + b^2}$

Ex.1 Find the equation of the parabola whose vertex is $(-3, 0)$ and directrix is $x + 5 = 0$.

Sol. A line passing through the vertex $(-3, 0)$ and perpendicular to directrix $x + 5 = 0$ is the axis of the parabola. Let focus of the parabola is $(a, 0)$. Since vertex, is the middle point of $(-5, 0)$ and focus S, therefore

$$-3 = \frac{(a-5)}{2} \Rightarrow a = -1$$

$\therefore$ Focus = $(-1, 0)$

Thus the equation to the parabola is

$$(x + 1)^2 + y^2 = (x + 5)^2 \Rightarrow y^2 = 8(x + 3)$$

Ex.2 Find the vertex of the parabola $x^2 - 8y - x + 19 = 0$.

Sol. Given equation can be written as

$$\left(x - \frac{1}{2}\right)^2 - 8y + 19 - \frac{1}{4} = 0$$

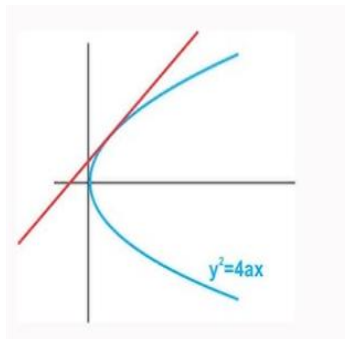
$$\Rightarrow \left(x - \frac{1}{2}\right)^2 = 8y - \frac{76-1}{4} \Rightarrow \left(x - \frac{1}{2}\right)^2 = 8\left(y - \frac{75}{32}\right)$$

$$\therefore \text{vertex} = \left(\frac{1}{2}, \frac{75}{32}\right)$$

Equation of a tangent to the Parabola

The equation of the tangent to a parabola depends on the form of the parabola. For the standard parabola $y^2 = 4ax$, the tangent equation can be written in slope form as $y = mx + \frac{a}{m}$ and

in point form as $yy_1 = 2a(x + x_1)$ at (x_1, y_1) .



~~Ex~~ Reduce the equation —

$$16x^2 - 24xy + 9y^2 - 104x - 172y - 44 = 0 \quad \dots \quad (1)$$

to the standard form. Find all its properties.

As the second degree terms of (i) form a perfect square the eq.

(i) will represent a parabola since $\Delta \neq 0$ and $ab = h^2$

$$\therefore (4x - 3y)^2 = 104x + 172y - 44 \quad \dots \quad (2)$$

The lines $4x - 3y = 0$ and $104x + 172y - 44 = 0$ are not at right angles, hence let us introduce a constant λ in (2)

$$(4x - 3y + \lambda)^2 = (104x + 172y - 44) + 2\lambda(4x - 3y) + \lambda^2$$

$$\text{or, } (4x - 3y + \lambda)^2 = x(104 + 8\lambda) + y(172 - 6\lambda) + (\lambda^2 - 44)$$

$$\text{The lines } 4x - 3y + \lambda = 0 \quad \dots \quad (4)$$

$$\text{and } x(104 + 8\lambda) + y(172 - 6\lambda) + (\lambda^2 - 44) = 0 \quad \dots \quad (5)$$

are at right angles if (apply $a_1a_2 + b_1b_2 = 0$, sec Art. (25b))

$$4(104 + 8\lambda) + (-3)(172 - 6\lambda) = 0 \text{ or, } \lambda = 2 \quad \dots \quad (6)$$

Now eq. (3) becomes by (6)

$$(4x - 3y + 2)^2 = 40(3x + 4y - 1)$$

which can be written as

$$\left(\frac{4x - 3y + 2}{\sqrt{4^2 + 3^2}} \right)^2 \cdot (25) = 40 \cdot 5 \left(\frac{3x + 4y - 1}{\sqrt{3^2 + 4^2}} \right)$$

$$\text{or, } PN^2 = \frac{40 \cdot 5}{25} PM \text{ or, } Y^2 = 8X$$

Which is the standard form of eq. (1)

$$\text{or, } Y^2 = 4PX. \text{ Where } P = \frac{1}{4} \cdot 8 = 2$$

$$\text{For other properties of } Y^2 = 8x = 4pX \quad \dots \quad (7)$$

$$\text{Here } Y = \frac{4x - 3y + 2}{\sqrt{4^2 + 3^2}} \text{ and } X = \frac{3x + 4y - 1}{\sqrt{3^2 + 4^2}} \dots \dots \dots (8)$$

$$(i) \text{ Axis of (7) is } Y = 0 \text{ or, } 4x - 3y + 2 = 0, \text{ by (8)} \dots \dots (9)$$

$$(ii) \text{ Tangent at vertex is } X = 0, \text{ or, } 3x + 4y - 1 = 0 \text{ by (8)} \dots \dots (10)$$

(iii) Vertex is the intersection of eq. (9) and (10) by (8)

i. e. $4x - 3y + 2 = 0$ and $3x + 4y - 1 = 0$, Solve.

$$x = -\frac{1}{5}, y = \frac{2}{5} \text{ Vertex } \left(-\frac{1}{5}, \frac{2}{5} \right)$$

(iv) Latus rectum is $4p = 8$

(v) Focus $X = p, Y = 0$, by (8)

$$\text{or, } \frac{3x + 4y - 1}{\sqrt{3^2 + 4^2}} = 2 \text{ and } \frac{4x - 3y + 2}{\sqrt{4^2 + 3^2}} = 0$$

or, $3x + 4y - 11 = 0$ and $4x - 3y + 2 = 0$, Solve

$\therefore x = 1, y = 2$. The focus is $(1, 2)$

(vi) Foot of the directrix. $X = -p$ and $Y = 0$ by (8)

$$\text{or, } \frac{3x + 4y - 1}{\sqrt{3^2 + 4^2}} = -2 \text{ and } \frac{4x - 3y + 2}{\sqrt{4^2 + 3^2}} = 0$$

or, $3x + 4y + 9 = 0$ and $4x - 3y + 2 = 0$. Now solve. The foot of the directrix is $(-7/5, -6/5)$

(vii) Equation of the directrix. $X = -p$ by (8)

$$\text{or, } \frac{3x + 4y - 1}{\sqrt{3^2 + 4^2}} = -2 \text{ or, } 3x + 4y + 9 = 0$$

(viii) Equation of the latus rectum $X = p$ by (8)

$$\text{or, } \frac{3x + 4y - 1}{\sqrt{3^2 + 4^2}} = 2 \text{ or, } 3x + 4y - 11 = 0$$

Ex 9. Reduce the equation

$$4x^2 - 4xy + y^2 - 8x - y + 6 = 0$$

to the standard form. Find all its properties.

In the eq. (1), $a = 4, b = 1, h = -2, g = -4, f = -3, c = 5$

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = -64, ab - h^2 = 4 - 4 = 0$$

$\Delta \neq 0$ and $ab = h^2$. Hence eq. (1) represents a parabola. The eq. (1) may be written as

$$(2x - y)^2 = 8x + 6y - 5 \quad \dots \quad (2)$$

$$\text{or, } (2x - y + \lambda)^2 = 8x + 6y - 5 + \lambda^2 + 2\lambda(2x - y) \quad \dots \quad (3)$$

$$\text{or, } (2x - y + \lambda)^2 = x(4\lambda + 8) + y(-2\lambda + 6) + (\lambda^2 - 5) \quad \dots \quad (4)$$

$$\text{The lines } 2x - y + \lambda = 0 \quad \dots \quad (5)$$

$$\text{and } x(4\lambda + 8) + y(-2\lambda + 6) + (\lambda^2 - 5) = 0 \quad \dots$$

may be perpendicular if $a_1a_2 + b_1b_2 = 0$

$$\text{i. e. } 2(4\lambda + 8) + (-1)(-2\lambda + 6) = 0 \text{ or, } \lambda = -1 \quad \dots \quad (6)$$

The eq. (3) becomes

$$(2x - y - 1)^2 = 4(x + 2y - 1) \text{ which can be written as}$$

$$\left(\frac{2x - y - 1}{\sqrt{2^2 + 1^2}} \right)^2 = 4\sqrt{5} \left(\frac{x + 2y - 1}{\sqrt{1^2 + 2^2}} \right)$$

It is of the form $PN^2 = 4(\sqrt{5})/5$. PM.

or, $Y^2 = 4/\sqrt{5}$. X. which is the standard equation of a parabola.

$$(i) \text{ Latus rectum} = 4p = 4/\sqrt{5} \text{ or, } p = 1/\sqrt{5}$$

$$X = (x + 2y - 1)/\sqrt{5} \text{ and } Y = (2x - y - 1)/\sqrt{5}$$

$$\therefore Y^2 = 4pX.$$

$$(ii) \text{ Axis is } Y = 0 \text{ or, } 2x - y - 1 = 0 \text{ from (7) } \dots$$

$$(iii) \text{ Tangent at the vertex } X = 0 \text{ i. e. } x + 2y - 1 = 0 \text{ from (7) } \dots$$

(iv) Vertex A is the intersection of eqs. (8) and (9).

$$\text{Solve } x = 3/5, y = 1/5; A(3/5, 1/5)$$

$$(v) \text{ Focus is } S(p, 0) \text{ i.e. ; } X = p \text{ and } Y = 0$$

$$\text{or, } \frac{x + 2y - 1}{\sqrt{5}} = \frac{1}{\sqrt{5}} \text{ and } \frac{2x - y - 1}{\sqrt{5}} = 0 \text{ from (7)}$$

$$\text{or, } x + 2y - 2 = 0 \text{ and } 2x - y - 1 = 0$$

$$\text{Solve, } x = 4/5, y = 3/5; S(4/5, 3/5)$$

(vi) Equation of latus rectum is $X = p$
or, $(x + 2y - 1)/\sqrt{5} = 1/\sqrt{5}$ or, $x + 2y - 2 = 0$

(vii) Foot of the directrix $Z(-p, 0)$ i. e. $X = -p, Y = 0$
or, $(x + 2y - 1)/\sqrt{5} = -1/\sqrt{5}$ and $(2x - y - 1)/\sqrt{5} = 0$
Solve, $x = 4/5, y = -1/5 \therefore Z(4/5, -1/5)$.

(viii) Equation of the directrix is $X = -p$
or, $(x + 2y - 1)/\sqrt{5} = -1/\sqrt{5}$ or, $x + 2y = 0$

Exercise:

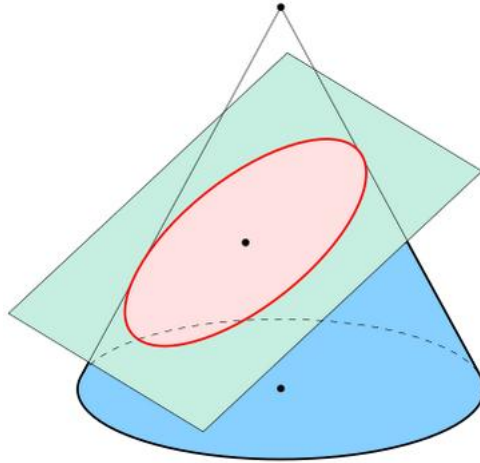
1. Find the nature of the conic $x^2 + 2xy + y^2 - 6x - 2y + 4 = 0$. Reduce it to standard form. Find vertex, focus, equation and length of the latus rectum.

Ans: $\left(\frac{x+y-2}{\sqrt{2}}\right)^2 = 4 \cdot \frac{\sqrt{2}}{4} \left(\frac{x-y}{\sqrt{2}}\right), \text{vertex: } (1,1), \text{focus: } \left(\frac{5}{4}, \frac{3}{4}\right),$

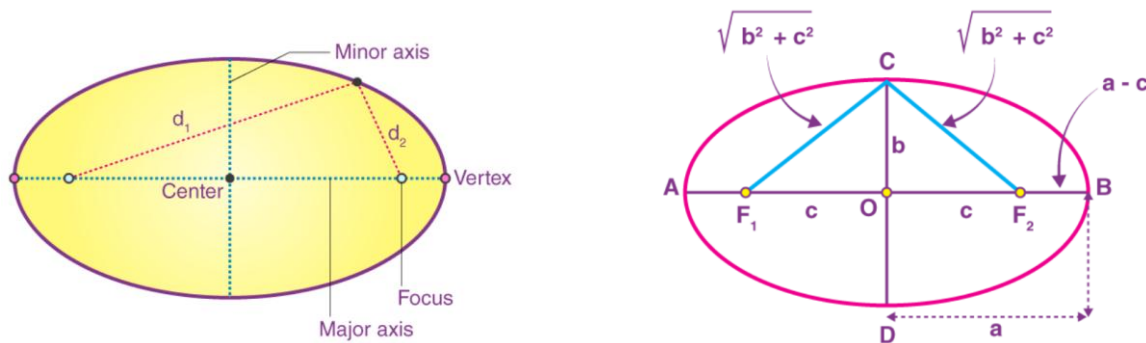
equ. of latus rectum, $2x - 2y = 1$, length of latus rectum $= \sqrt{2}$.

Ellipse

An ellipse is a closed, oval-shaped curve where the sum of the distances from any point on the curve to two fixed points (foci) is constant.



The ellipse is one of the conic sections, that is produced, when a plane cuts the cone at an angle with the base. If the cone is intersected by the plane, parallel to the base, then it forms a circle.



When the center of the ellipse is at the origin (0,0) and the foci are on the x-axis and y-axis, then

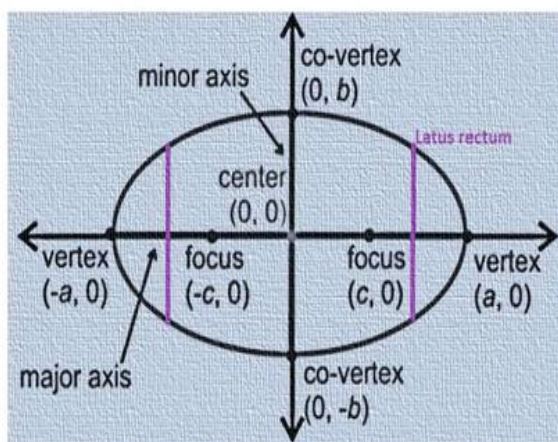
The equation of the ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

The eccentricity of an ellipse, $e = \frac{c}{a} = \sqrt{1 - \left(\frac{b^2}{a^2}\right)}$

The eccentricity of an ellipse lies between 0 to 1. $0 < e < 1$.

The length of the latus rectum for an ellipse is $2b^2/a$, if the major axis is along the x-axis.

$$L = \frac{2 \times (\text{semi-minor axis})^2}{\text{semi-major axis}}$$



The standard form of the equation of an ellipse with center $(0, 0)$ is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

where

- $a > b$
- the length of the major axis is $2a$
- the coordinates of the vertices are $(\pm a, 0)$
- the length of the minor axis is $2b$
- the coordinates of the co-vertices are $(0, \pm b)$
- the coordinates of the foci are $(\pm c, 0)$, where $c^2 = a^2 - b^2$

Ellipse Equation

Area of ellipse = πab

$$\text{Perimeter of ellipse} = 2\pi \sqrt{\frac{a^2 + b^2}{2}}$$

The standard form of the equation of an ellipse with center (h, k) is

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

where

- $a > b$
- the length of the major axis is $2a$
- the coordinates of the vertices are $(h \pm a, k)$
- the length of the minor axis is $2b$
- the coordinates of the co-vertices are $(h, k \pm b)$
- the coordinates of the foci are $(h \pm c, k)$, where $c^2 = a^2 - b^2$

Reduce to standard form (Ellipse)

Working Rule : For the reduction of the equation of a conic referred to centre as origin.
Let the equation of the conic be.

$$F(x, y) \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

$$\text{then write } \frac{\delta F}{\delta x} \equiv (ax + hy + g) = 0 \text{ or } ax + hy + g = 0$$

$$\text{and } \frac{\delta F}{\delta y} = 2(hx + by + f) = 0 \text{ or } hx + by + f = 0$$

$$\text{The equations (2) and (3) i. e. } \frac{\delta F}{\delta x} = 0, \text{ and } \frac{\delta F}{\delta y} = 0$$

are the same as these giving the co-ordinates of the centre.

Now the constant $= \frac{\Delta}{ab - h^2} = gx + fy + c$ which is represented by another constant c_1 .

$$c_1 = gx + fy + c$$

where x and y are found from the equations $\frac{\delta F}{\delta x} = 0$, and $\frac{\delta F}{\delta y} = 0$... (4)

and g, f and c are found from the given equation (1)

Hence the equation reduces to $ax^2 + 2hxy + by^2 + c_1 = 0$.

In other words "Same terms of second degree as in the given equation + new constant = 0.
which we can always reduce to the FORM

$$Ax^2 + 2Hxy + By^2 = 1$$

... (5)

If the axes of the equation (5) are rotated about its new (centre) origin in such a way that xy -term vanishes, the new equation of the conic becomes $a_1x^2 + b_1y^2 + c_1 = 0$... (6)

which is called the Standard Form of the equation of the conic. Then by Invariants, (See Art. 35) we have from (5) and (6)

$$a_1 + b_1 = a + b$$

$$a_1b_1 = ab - h^2$$

where a_1 and b_1 can be easily found.

Thus the equation (1) reduces to the standard form

$$a_1x^2 + b_1y^2 + c_1 = 0$$

Ex. 14(a) What conic does the equation $5x^2 - 2xy + 5y^2 - 8x - 8y - 8 = 0$ represent? Reduce the equation to standard form. [N.U.H. 1996]

Ans. The given equation

$$5x^2 - 2xy + 5x^2 - 8x - 8y - 8 = 0 \quad \dots \quad (i)$$

Comparing (i) with $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

We get, $a = 5, h = -1, b = 5, g = -4, f = -4, c = -8$.

$$\therefore \Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$$= 5 \cdot 5 \cdot (-8) + 2(-4)(-4)(-1) - 5(-4)^2 - 5(-4)^2 - (-8)(-1)^2$$

$$= -200 - 32 - 80 - 80 + 8 = -384$$

$$\therefore \Delta \neq 0$$

$$ab - h^2 = 5 \cdot 5 - (-1)^2 = 25 - 1 = 24$$

$$\therefore ab - h^2 > 0$$

Thus, the given conic represents an ellipse,

Let (α, β) be the centre of the conic (i)

$$\therefore \alpha = \frac{hf - bg}{ab - h^2}, \quad \beta = \frac{hg - af}{ab - h^2}$$

$$= \frac{(-1)(-4) - 5(-4)}{5.5 - (-1)^2}$$

$$= \frac{24}{24}$$

$$= 1$$

$$= \frac{(-1)(-4) - 5(-4)}{5.5 - (-1)^2}$$

$$= \frac{24}{24}$$

$$= 1$$

∴ Centre is at (1, 1)

Transfer the origin to a point (1,1) When direction of axes are unchanged, then (1) becomes

$$ax^2 + 2hxy + by^2 = c_1$$

$$\text{ie } 5x^2 - 2xy + 5y^2 = c_1$$

$$\text{where } c_1 = -(\alpha g + \beta f + c)$$

$$c_1 = -\{1 \cdot (-4) + 1 \cdot (-4) - 8\} = 16$$

$$\therefore 5x^2 - 2xy + 5y^2 = 16$$

$$\Rightarrow \frac{5}{16}x^2 - 2 \cdot \frac{1}{16}xy + \frac{5}{16}y^2 = 1$$

$$\Rightarrow Ax^2 + 2Hxy + By^2 = 1 \dots \dots (2)$$

$$\text{Where } A = \frac{5}{16}, H = \frac{-1}{16}, B = \frac{5}{16}$$

Removing xy term when the axes being rotated.

$$\therefore a_1x^2 + b_1y^2 = 16 \dots \dots (3)$$

$$a_1 + b_1 = a + b$$

$$a_1b_1 = ab - h^2$$

$$\therefore a_1 + b_1 = 5 + 5 = 10$$

$$a_1b_1 = 5.5 - (-1)^2$$

$$a_1b_1 = 24$$

$$\text{Now, } (a_1 - b_1)^2 = (a_1 + b_1)^2 - 4a_1b_1 \\ = 10^2 - 4 \cdot 24 = 100 - 96 = 4$$

$$\therefore a_1 - b_1 = 2 \text{ take +ve sign}$$

$$\therefore a_1 + b_1 = 10$$

$$a_1 - b_1 = 2$$

$$(+)\quad 2a_1 = 12$$

$$b_1 = 10 - a_1 = 10 - 6 = 4$$

$$\therefore (3) \Rightarrow 6x^2 + 4y^2 = 16$$

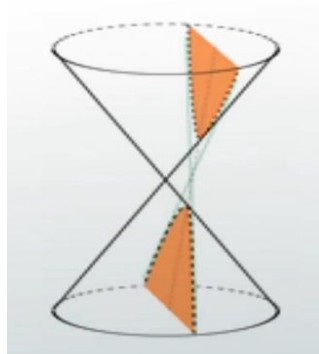
$$\Rightarrow \frac{x^2}{\frac{8}{3}} + \frac{y^2}{4} = 1$$

$$\Rightarrow \frac{x^2}{\left(\sqrt{\frac{8}{3}}\right)^2} + \frac{y^2}{y^2} = 1$$

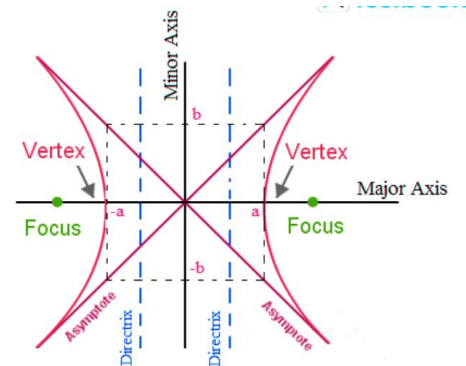
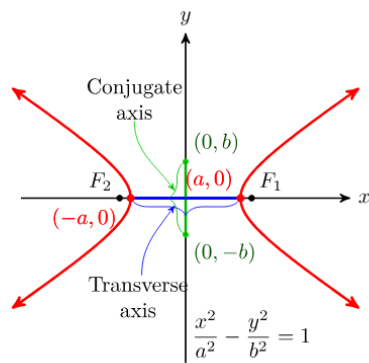
which is standard form of an ellipse.

Hyperbola

A hyperbola is a special type of conic section formed when a plane cuts through both halves of a double cone, but not exactly at the center.



A hyperbola is the set of points in the plane, the difference of whose distances from two fixed points (the foci) is a constant.



Standard equation of the hyperbola with center (h, k) and the X-axis as the transverse axis and the Y-axis as the conjugate axis is,

$$\frac{(x - h)^2}{a^2} - \frac{(y - k)^2}{b^2} = 1$$

The equation of the ellipse with center at (0,0) is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Eccentricity

The eccentricity of a hyperbola is the ratio of the distance of a point from the focus to its perpendicular distance from the directrix. The eccentricity of a hyperbola is always greater than 1, i.e., $e > 1$. $e = \sqrt{1 + (b^2/a^2)}$

Properties

Forms of the hyperbola	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$
Coordinates of centre	(0, 0)	(0, 0)
Coordinates of vertices	($\pm a$, 0)	(0, $\pm a$)
Coordinates of foci	($\pm ae$, 0)	(0, $\pm ae$)
Length of transverse axis	2a	2a
Length of conjugate axis	2b	2b
Equation of directrices	$x = \pm a/e$	$y = \pm a/e$
Eccentricity	$e = \sqrt{\frac{a^2 + b^2}{a^2}}$	$e = \sqrt{\frac{a^2 + b^2}{a^2}}$
Length of latusrectum	$2b^2/a$	$2b^2/a$
Equation of transverse axis	$y = 0$	$x = 0$
Equation of conjugate axis	$x = 0$	$y = 0$

Ex. 14(c) Reduce the equation $x^2 + 12xy - 4y^2 - 6x + 4y + 9 = 0$ to the standard form and identify it. [N.U.H. 1997]

Ans. The given equation

$$x^2 + 12xy - 4y^2 - 6x + 4y + 9 = 0 \dots (1)$$

Compare (1) with the equation

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \text{ We get}$$

$$a = 1, b = -4, h = 6, g = -3, f = 2, c = 9$$

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$$= 1 \cdot (-4) \cdot 9 + 2 \cdot 2 \cdot (-3) \cdot 6 - 1 \cdot 2^2 + 4 \cdot (-3)^2 - 9 \cdot 6^2$$

$$= -36 - 72 - 4 + 36 - 324$$

$$= -400 \neq 0$$

$$ab - h^2 = 1 \cdot -4 - 6^2 = -40 < 0$$

Thus, the given conic represents a hyperbola.

Let (α, β) be the centre of the conic (1)

$$\begin{aligned}\therefore \alpha &= \frac{hf - bg}{ab - h^2} & \beta &= \frac{hg - af}{ab - h^2} \\&= \frac{6 \cdot 2 + 4(-3)}{1 \cdot 4 - 6^2} & &= \frac{6 \cdot 3 - 1 \cdot 2}{1 \cdot 4 - 6^2} \\&= \frac{0}{-40} & &= \frac{-20}{-40} \\&= 0 & &= \frac{1}{2}\end{aligned}$$

$\therefore$ Centre is at $(0, \frac{1}{2})$

Now, transfer the origine to a point $(0, \frac{1}{2})$

Then (1) becomes

$$x^2 + 12xy - 4y^2 = c_1 \dots\dots (2)$$

Where $C_1 = -(g\alpha + f\beta + C)$

$$= -(-3 \cdot 0 + 2 \cdot \frac{1}{2} + 9)$$

$$= -10$$

$$\Rightarrow x^2 + 12xy - 4y^2 = -10 \dots\dots(3)$$

Now, transform the axes such that remove the xy term, then

$$a_1 x^2 + b_1 y^2 = -10 \dots\dots(4)$$

$$\text{Where } a_1 b_1 = a + b = 1 - 4 = -3$$

$$\text{and } a_1 b_1 = ab - h^2 = 4 - 6^2 = -40$$

$$\begin{aligned}\text{Now, } (a_1 - b_1)^2 &= (a_1 + b_1)^2 - 4 a_1 b_1 \\&= (-3)^2 - 4 \cdot -40 \\&= 9 + 160 = 169\end{aligned}$$

$$\therefore a_1 - b_1 = 13 \quad \text{Take + ve sign}$$

$$\text{Again, } a_1 + b_1 = 3$$

$$\frac{a_1 - b_1 = 13}{(+)\ 2a_1 = 10} \quad \therefore a_1 = 5$$

$$b_1 = -3 - a_1 = -3 - 5 = -8$$

$$(4) \Rightarrow 5x^2 - 8y^2 = -10$$

$$\Rightarrow \frac{x^2}{-2} + \frac{y^2}{\frac{5}{4}} = 1$$

Which is required standard form.

Ex. 14(b) Find the asymptotes the hyperbola $x^2 + 2xy - y^2 + 2x + 4y = 0$ and the equation of the conjugate hyperbola. [N.U.H. 1996]

Ans. The given hyperbola

$$x^2 + 2xy - y^2 + 2x + 4y = 0 \dots (1)$$

Let equation of asymptote of the hyperbola (1) is

$$x^2 + 2xy - y^2 + 2x + 4y + \lambda = 0 \dots (2)$$

Since asymptotes is a pair of st. lines

$$\text{So } \Delta = 0 \text{ is } \boxed{abc + 2fgh - af^2 - bg^2 - ch^2 = 0}$$

$$\therefore 1 \cdot (-1) \cdot \lambda + 2 \cdot (+2) \cdot (+1) \cdot (-1) \cdot (+2)^2 - (-1) \cdot (+1)^2 - \lambda \cdot 1^2 = 0$$

$$\Rightarrow -\lambda + 4 - 4 + 1 - \lambda = 0$$

$$\Rightarrow 2\lambda + 1 = 0$$

$$\therefore \lambda = -\frac{1}{2}$$

$$\therefore (2) \Rightarrow x^2 + 2xy - y^2 + 2x + 4y + \frac{1}{2} = 0$$

Which is the required asymptotes.

Let equation of the conjugate hyperbola is

$$x^2 + 2xy - y^2 + 2x + 4y + k = 0 \dots (3)$$

We know, equation of the conjugate hyperbola

$$= 2 \times \{\text{equation of asymptotes}\} - \text{equation of the hyperbola}$$

$$\Rightarrow x^2 + 2xy - y^2 + 2x + 4y + k = 2 \times (x^2 + 2xy - y^2 + 2x + 4y + \frac{1}{2}) - (x^2 + 2xy - y^2 + 2x + 4y)$$

$$\Rightarrow k = 1$$

$$(3) \Rightarrow x^2 + 2xy - y^2 + 2x + 4y + 1 = 0$$

Which is the required conjugate hyperbola.

Exercise

1. Reduce the equation $x^2 + 2xy + y^2 - 2x - 1 = 0$ to its standard form and then identify it. Find the coordinates of its focus, the equation of the axis and the directrix.

$$\text{Ans: Parabola; } \left(\frac{x+y-\frac{1}{2}}{\sqrt{2}}\right)^2 = 4 \cdot \frac{1}{4\sqrt{2}} \cdot \frac{x-y+\frac{5}{2}}{\sqrt{2}}; (-1/4, 3/4); 2x+2y-1=0; 2x-2y+3=0.$$

2. Reduce the equation $8x^2 + 4xy + 5y^2 - 24x + 24y = 0$ to its standard form and then identify it. Find the equation of the axes and eccentricity.

$$\text{Ans: Ellipse; } \frac{x^2}{98/27} + \frac{y^2}{98/12} = 1.$$

3. Reduce the equation $x^2 - 5xy + y^2 + 8x - 20y + 15 = 0$ to its standard form and then identify it. Find the equation of the axes and its latus rectum, eccentricity.

Ans: Hyperbola; $\frac{x^2}{2/7} - \frac{y^2}{\frac{2}{3}} = 1$; equ. of transverse axes, $x+y+4=0$; equ. of conjugate axes, $x-y+4$; $e=\sqrt{10/3}$; latus rectum=4.

General Equation of Second Degree

15. Find the centre of the following conics

(i) $x^2 + y^2 + xy + x + y - 1 = 0$

Ans. $(-1/3, -1/3)$

(ii) $17x^2 - 12xy + 8y^2 + 46x - 28y + 17 = 0$

Ans. $(-1, 1)$

(iii) $3x^2 - 5xy + 6y^2 + 11x - 17y + 13 = 0$

Ans. $(-1, 1)$

16. What conics do the equations represent?

(i) $2x^2 + 3xy - 2y^2 - 5x + 5y = 0$ [E.U. 1961] Ans. Rect hyperbola.

(ii) $7x^2 - 6xy - y^2 + 4x - 4y = 0$

Ans. Pair of straight lines

17. Find the lengths, equations and directions of axes of the following conics.

(i) $17x^2 + 12xy + 8y^2 - 46x - 28y + 33 = 0$

Ans. $4/\sqrt{5}, 2/\sqrt{5}$

$2x + y - 3 = 0, x - 2y + 1 = 0, \tan^{-1}(-2), \tan^{-1}\frac{1}{2}$

[C.U.(S) 1992]

(ii) $x^2 + 4xy + y^2 - 2x + 2y - 6 = 0$

Ans. $4/\sqrt{3}, x - y + 2 = 0, x + y = 0, 45^\circ, 135^\circ$

18. Find the eccentricity of the conics (কণিকের বিকেন্দ্রিকতা)

(a) $34x^2 + 24xy + 41y^2 + 48x + 14y - 108 = 0$; $(-\frac{16}{26}, \frac{1}{25})$

Ans. $1/\sqrt{2}$

(b) $x^2 + 4xy + y^2 - 2x + 2y - 6 = 0$

Ans. 2

19. Find the focus (নাভি) of the conics

(a) $9x^2 + 24xy + 16y^2 - 2x + 14y + 1 = 0$

Ans. $[-1/25, -11/50]$

(b) $x^2 + 4xy + y^2 - 2x + 2y - 6 = 0$

Ans. $[-1 \pm 2\sqrt{2}/3, 1 \pm 2\sqrt{2}/3]$

(c) $4x^2 + y^2 - 4xy - 10y - 19 = 0$

Ans. $[-6/5, -7/5]$

(d) $20x^2 - 4xy + 5y^2 - 16x - 14y + 17 = 0$

Ans. $[2, 3], [1, 1]$

20. Find the directrix (নিয়ামক) of the conics

(a) $8x^2 - 4xy + 5y^2 - 16x - 14y + 17 = 0$

Ans. $x + 2y - 10 = 0, x + 2y - 1 = 0$

(b) $9x^2 - 6xy + y^2 - 14x - 2y - 12 = 0$

Ans. $2x + 6y - 7 = 0$ [C.U. 1979, D.U. 1979]

21. Calculate the latus rectum (নাভিসংখ্য) of the conic

$25x^2 + 120xy + 144y^2 - 82x - 163y + 32 = 0$

Ans. $1/13$

22. Find the co-ordinates of vertex, focus and equation of the latus rectum of the conic $x^2 + y^2 - 6x - 2y + 4 = 0$ $\lambda = -2$. [N.U.H. 99] Ans. $(1, 1); (5/4, 3/4); 2x - 2y - 1 = 0$

23. (a) Find the equations of latus rectum and directrix of the conic $55x^2 - 30xy + 39y^2 - 24x - 24y - 464 = 0$

Ans. $3x + 5y - 21 = 0, 3x + 5y + 13 = 0$

Ans. $(1/2, 1/2), c_1 = -480, 5x + 5y = 36, 3x + 5y + 28 = 0$

24. Find the vertex, directrix and focus of the conics

(a) $9x^2 - 6xy + y^2 - 14x - 2y + 12 = 0$

Ans. $(1, 1); 2x + 6y - 7 = 0; (21/20, 23/20)$

(b) $x^2 - 4xy + 4y^2 + 10x - 8y + 13 = 0$

Ans. $(-39/25, 13/25), 10x + 5y - 2 = 0, (-9/5, 2/5)$

25. Find the focus and directrix of the conic

$17x^2 - 12xy + 8y^2 + 46x - 28y + 17 = 0$

Ans. $(-1 \pm \sqrt{3/5}, 1 \pm \sqrt{3/5}); x + 2y = 1 \pm 120/\sqrt{15}$

26. Find the tangents at the vertex (শীর্ষ স্পর্শক) of the conic

$11x^2 + 16xy - y^2 - 70x - 40y + 82 = 0$

Ans. $x - y = -1 \pm 2\sqrt{5/6}$